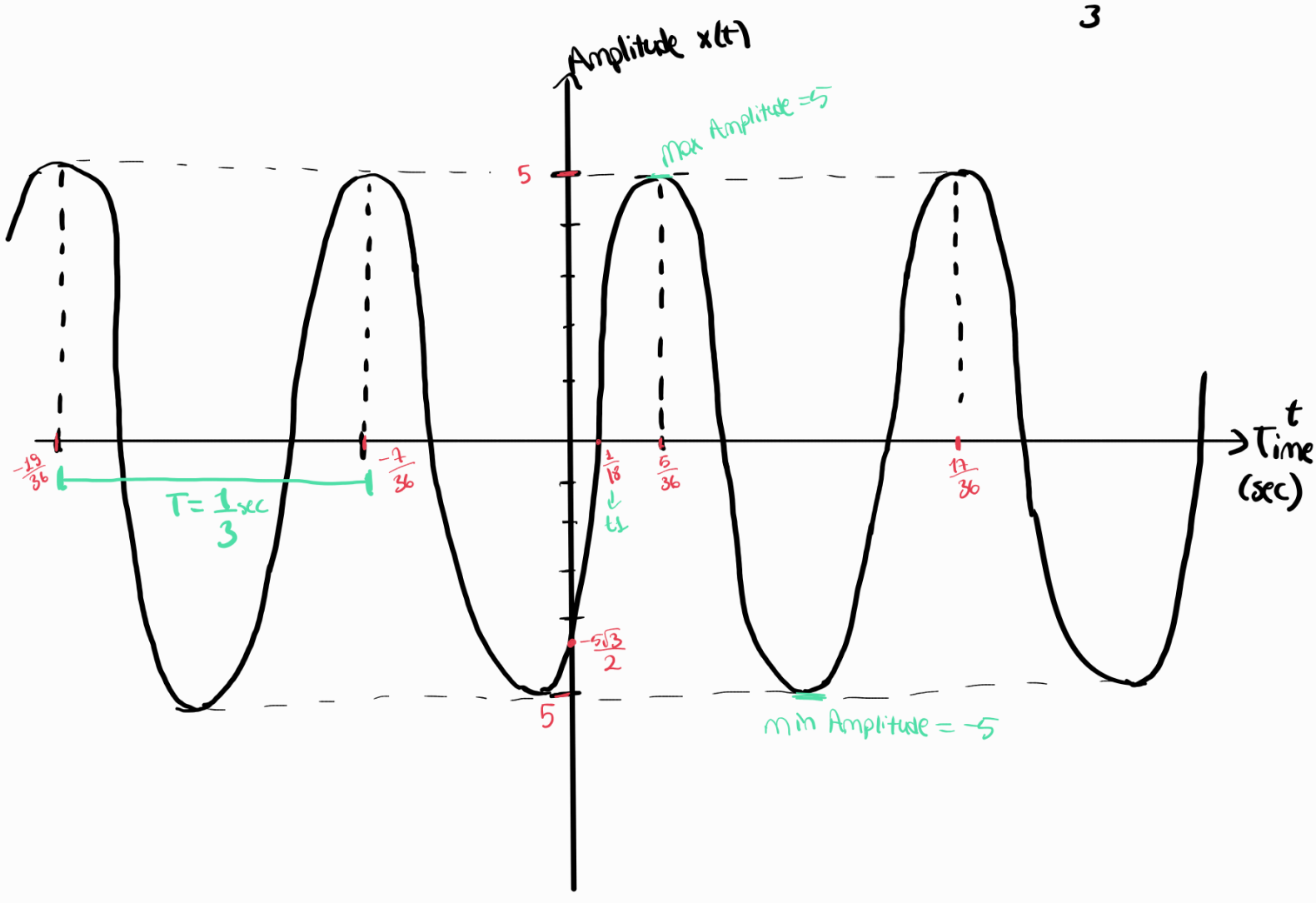
1) Plotting the sinusoids:

a) $x(t) = 5\sin(6\pi t - \frac{\pi}{3})$, $A = 5$, $\omega_0 = 6\pi = \frac{2\pi}{T_0} \Rightarrow T_0 = \frac{1}{3}$ sec

$x(0) = 5\sin(-\frac{\pi}{3}) = \frac{-5\sqrt{3}}{2} \Rightarrow -5 < x(0) < -4$, $t_1 = \frac{-\phi}{\omega_0} \Rightarrow t_1 = \frac{\frac{\pi}{3}}{6\pi} = \frac{1}{18}$ sec

$\phi = -\frac{\pi}{3}$



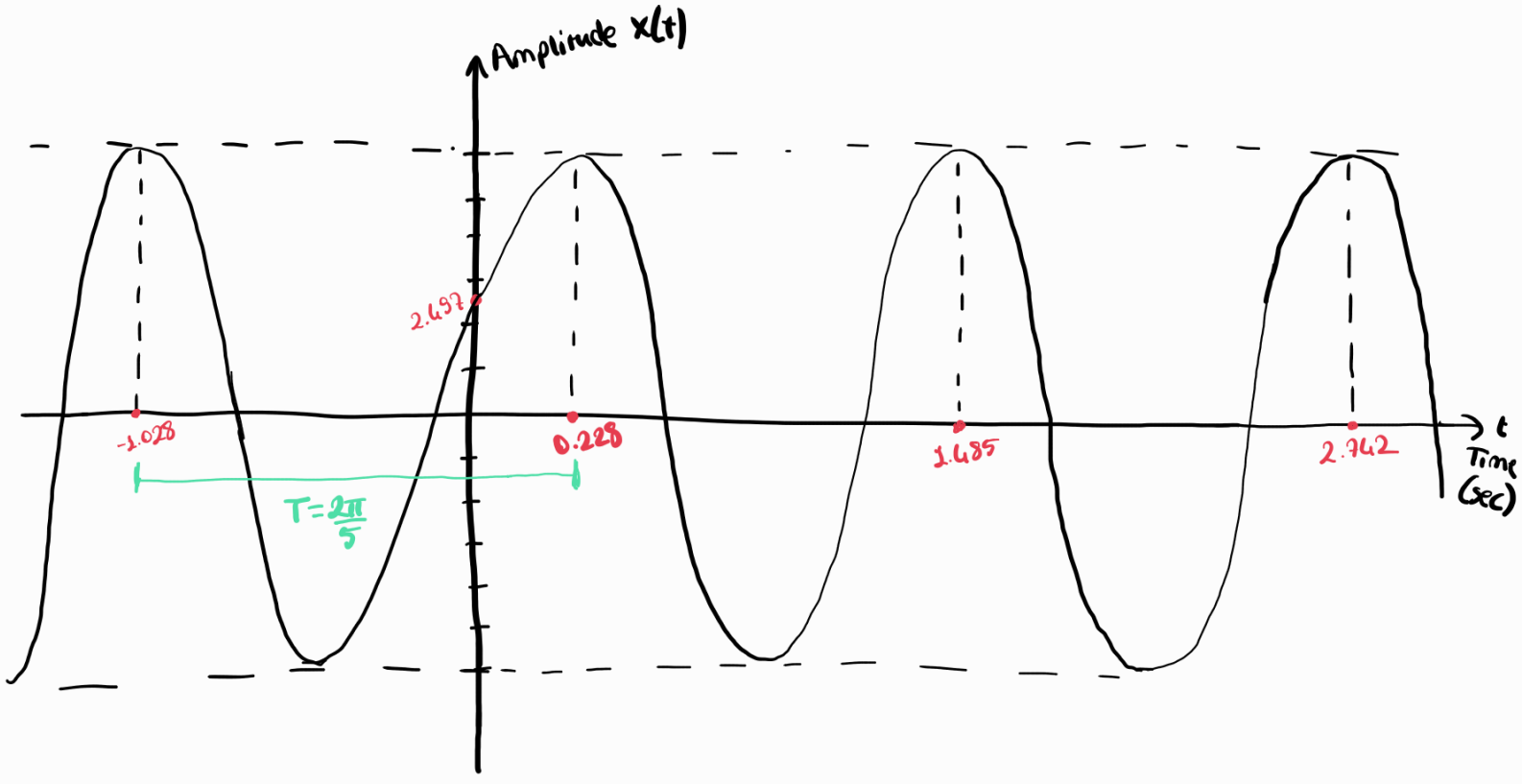
in this example, solving "$\omega_0 t_1 + \phi = 0$" gives the first intersection of the sinusoid with x-axis (to the left or to the right, depending on the sign). This is because this is a sin function.

in part b, however, solving the same equation gives not the x-axis intersection but the first max amplitude's x-axis before or after the origin, since it is a cosine function

b) $x(t) = -6\cos(5t+2) = 6\cos(5t+2+\pi)$, $A = 6$,

$\omega_0 = 5 = \frac{2\pi}{T_0} \Rightarrow T_0 = \frac{2\pi}{5} \approx 1.257$, $x(0) = 6\cos(2+\pi) \approx 2.497$

$t_1 = \frac{-\phi}{\omega_0} \Rightarrow t_1 = \frac{-(2+\pi)}{5} \approx -1.028$ (minus sign means its shifted to left)



2) a) $x(t) = 3\sqrt{2}\cos\left(2t - \frac{\pi}{3}\right)$

$$\Rightarrow X = 3\sqrt{2}\, e^{-j\frac{\pi}{3}} = 3\sqrt{2}\cos\left(-\tfrac{\pi}{3}\right) + j\,3\sqrt{2}\sin\left(-\tfrac{\pi}{3}\right)$$

$$= 3\sqrt{2}\cos\left(\tfrac{\pi}{3}\right) - j\,3\sqrt{2}\sin\left(\tfrac{\pi}{3}\right) = \frac{3\sqrt{2}}{2} - \frac{j\,3\sqrt{6}}{2}$$

($\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$, $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$)

$$y(t) = 3\sin\left(2t + \tfrac{\pi}{4}\right)$$

$$\Rightarrow y = 3\, e^{j\frac{\pi}{4}} = 3\cos\left(\tfrac{\pi}{4}\right) + j3\sin\left(\tfrac{\pi}{4}\right) = \frac{3\sqrt{2}}{2} + \frac{j\,3\sqrt{2}}{2}$$

($\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$, $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$)

b) $Z = X + y \approx 4.243 - j\,1.553$

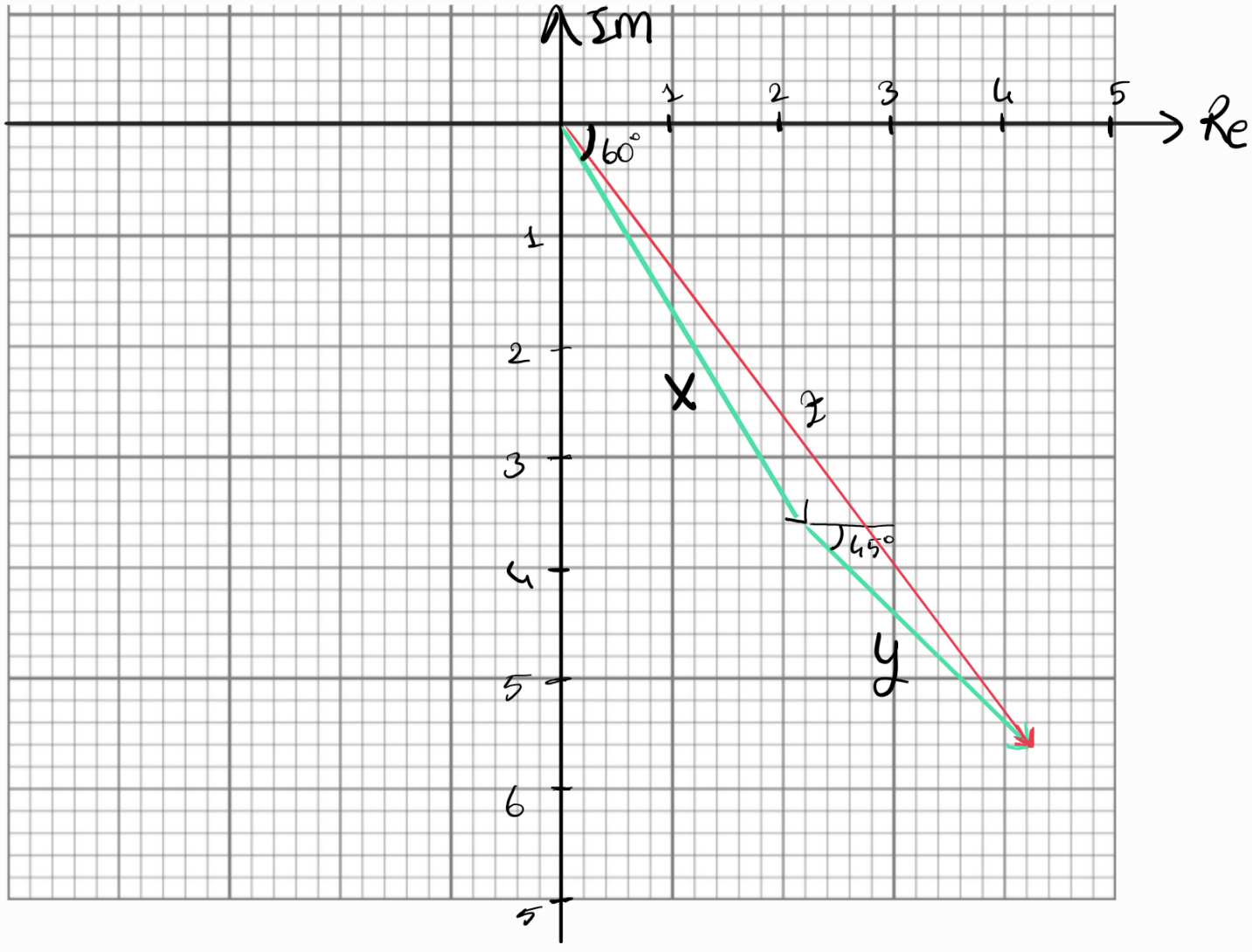
we will use Z to find $x(t) + y(t)$, which is in form $A\cos(2t + \phi)$:

$$A = \sqrt{Re\{Z\}^2 + Im\{Z\}^2} \Rightarrow A \approx 4.518$$

$$\phi = \arctan\left(\frac{Im\{Z\}}{Re\{Z\}}\right) \Rightarrow \phi \approx 0.351$$

$$\begin{pmatrix} Re\{Z\} = 4.243 \\ Im\{Z\} = -1.553 \end{pmatrix}$$

$$x(t) + y(t) = 4.518\cos(2t + 0.351)$$


Im
Re
1
2
3
4
5
60°
X
Z
45°
y
1
2
3
4
5
6

3) for a signal to be periodic, all sinusoids included must have frequencies that are integer multiples of fundamental frequency.

a) $x(t) = 3\sin(3t)\cos\left(\frac{\pi}{5}t + \frac{\pi}{3}\right) - 1$

since $\sin A\cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$;

$$x(t) = \frac{3}{2}\sin\left[\underbrace{\left(3+\frac{\pi}{5}\right)t + \frac{\pi}{3}}_{\omega_1}\right] + \frac{3}{2}\sin\left[\underbrace{\left(3-\frac{\pi}{5}\right)t + \frac{\pi}{3}}_{\omega_2}\right] - 1$$

since $f_1 = \frac{\omega_1}{2\pi}$, $f_2 = \frac{\omega_2}{2\pi}$ are not harmonically related: not periodic

b) $x(t) = 7\sin(\underbrace{\sqrt{3}t}_{\omega_1} + 5) + 2\cos(\underbrace{\pi t}_{\omega_2})$

same reason above, not periodic

c) $x(t) = \cos(\underbrace{12t}_{\omega_1}) - 3\cos(\underbrace{16t}_{\omega_2})$

since $f_1 = \frac{\omega_1}{2\pi} = \frac{6}{\pi}$, and $f_2 = \frac{\omega_2}{2\pi} = \frac{8}{\pi}$, we have

$\text{lcm}\left(\frac{6}{\pi}, \frac{8}{\pi}\right) = \boxed{\frac{2}{\pi} = f_0} \Rightarrow \boxed{T_0 = \frac{\pi}{2}}$ (periodic)

present harmonics are 3rd and 4th

d) $x(t) = \sin^2(60\sqrt{2}t) + \cos^2(24\sqrt{2}t + \frac{\pi}{4})$

since $\sin^2 A = \frac{(1-\cos 2A)}{2}$ and $\cos^2 A = \frac{(1+\cos 2A)}{2}$

$\Rightarrow x(t) = \frac{1}{2} - \frac{1}{2}\cos(120\sqrt{2}t) + \frac{1}{2} + \frac{1}{2}\cos(48\sqrt{2}t + \frac{\pi}{2})$

$\Rightarrow x(t) = 1 - \frac{1}{2}\cos(\underbrace{120\sqrt{2}t}_{\omega_1}) + \frac{1}{2}\cos(\underbrace{48\sqrt{2}t}_{\omega_2} + \frac{\pi}{2})$

$\text{lcm}(120\sqrt{2}, 48\sqrt{2}) = \omega_0 = 24\sqrt{2} \Rightarrow f_0 = \frac{24\sqrt{2}}{2\pi} = \frac{12\sqrt{2}}{\pi} \Rightarrow T_0 = \frac{\pi}{12\sqrt{2}}$

(periodic) 0th, 5th, and 2nd harmonics are present

4) $a_0 = -4$, $a_1 = -j = e^{j\frac{3\pi}{2}}$, $a_5 = -4e^{-j\frac{\pi}{3}} = 4e^{j\frac{2\pi}{3}}$ $a_8 = 2e^{j\frac{\pi}{3}}$

$$-j = \cos\frac{3\pi}{2} + j\sin\frac{3\pi}{2} = e^{j\frac{3\pi}{2}}$$

if $T_0 = 5$ sec, then $\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{5}$ rad/s, and $\omega_k = k \cdot \omega_0$

we can find (all are in rad/s units)

$\omega_{(0)} = 0 \cdot \frac{2\pi}{5} = 0$, $\omega_{(1)} = \frac{2\pi}{5}$, $\omega_{(5)} = 5 \cdot \frac{2\pi}{5} = 2\pi$, $\omega_{(8)} = 8 \cdot \frac{2\pi}{5} = \frac{16\pi}{5}$

and also we can say (since $A_k = 2a_k$, except A_0):

$A_0 = -4$, $A_1 = 2$, $A_5 = 8$, $A_8 = 4$

$\phi_0 = 0$, $\phi_1 = \frac{3\pi}{2}$, $\phi_5 = \frac{2\pi}{3}$, $\phi_8 = \frac{\pi}{3}$

$$\Rightarrow x(t) = -4 + 2\cos\left(\frac{2\pi}{5}t + \frac{3\pi}{2}\right) + 8\cos\left(2\pi t + \frac{2\pi}{3}\right) + 4\cos\left(\frac{16\pi}{5}t + \frac{\pi}{3}\right)$$

5)

$$x(t) = 5 - 2e^{jt} + 3\sin(-t + \pi/4) - 4\cos(3t+5) + \frac{3}{2}\cos(9t+\pi/2)$$

$$+ \frac{3}{2}\cos(t+\pi/2), \quad \left(\cos A\cos B = \frac{1}{2}\left[\cos(A+B)+\cos(A-B)\right]\right)$$

a) average $= a_0 = 5$

before continuing lets convert the sinusoids to complex form (since we will use it in part d)

$$3\sin(-t+\pi/4) = \frac{3}{2j}e^{-jt}\cdot e^{j\frac{\pi}{4}} - \frac{3}{2j}e^{jt}\cdot e^{-j\frac{\pi}{4}}$$

$$-4\cos(3t+5) = -2\cdot e^{j3t}\cdot e^{j5} - 2\cdot e^{-j3t}\cdot e^{-j5}$$

$$\frac{3}{2}\cos(9t+\pi/2) = \frac{3}{4}e^{j9t}e^{j\frac{\pi}{2}} + \frac{3}{4}e^{-j9t}\cdot e^{-j\frac{\pi}{2}}$$

$$\frac{3}{2}\cos(t+\pi/2) = \frac{3}{4}e^{jt}\cdot e^{j\frac{\pi}{2}} + \frac{3}{4}e^{-jt}e^{-j\frac{\pi}{2}}$$

rewriting $x(t) = 5 - 2e^{-jt} + \frac{3}{2j}e^{j\frac{\pi}{4}}e^{-jt} - \frac{3}{2j}e^{-j\frac{\pi}{4}}e^{jt}$

$$-2e^{j5}e^{j3t} - 2e^{-j5}e^{-j3t} + \frac{3}{4}e^{j\frac{\pi}{2}}e^{j9t} + \frac{3}{4}e^{-j\frac{\pi}{2}}e^{-j9t}$$

$$+\frac{3}{4}e^{j\frac{\pi}{2}}e^{jt} + \frac{3}{4}e^{-j\frac{\pi}{2}}e^{-jt}$$

b) we take the lcm of all ω's to get $\omega_0 = 1$ rad/s

then $T_0 = \frac{2\pi}{\omega_0} \Rightarrow \boxed{T_0 = 2\pi}$

c) No, $x(t)$ is not real because it's not symmetric due to $-2e^{-jt}$

d) some adjustments before plotting freq. spectrum:

$\frac{3}{2j} e^{j\frac{\pi}{4}} = \frac{3}{2} e^{j\frac{3\pi}{2}} e^{j\frac{\pi}{4}} = \frac{3}{2} e^{j\frac{7\pi}{4}}$ $\left(\frac{1}{j} = e^{j\frac{3\pi}{2}}\right)$

$\frac{-3}{2j} e^{-j\frac{\pi}{4}} = \frac{3}{2} e^{j\frac{\pi}{2}} e^{-j\frac{\pi}{4}} = \frac{3}{2} e^{j\frac{\pi}{4}}$ $\left(\frac{-1}{j} = j = e^{j\frac{\pi}{2}}\right)$

$-2e^{j5} = 2e^{j\pi} e^{j5} = 2e^{j(\pi+5)}$ $\left(-1 = e^{j\pi}\right)$

$-2e^{-j5} = 2e^{j\pi} e^{-j5} = 2e^{j(\pi-5)}$ $\left(-1 = e^{j\pi}\right)$

$a_0 = 5$

$a_1 = \frac{3}{2} e^{j\frac{\pi}{4}} + \frac{3}{4} e^{j\frac{\pi}{2}}$

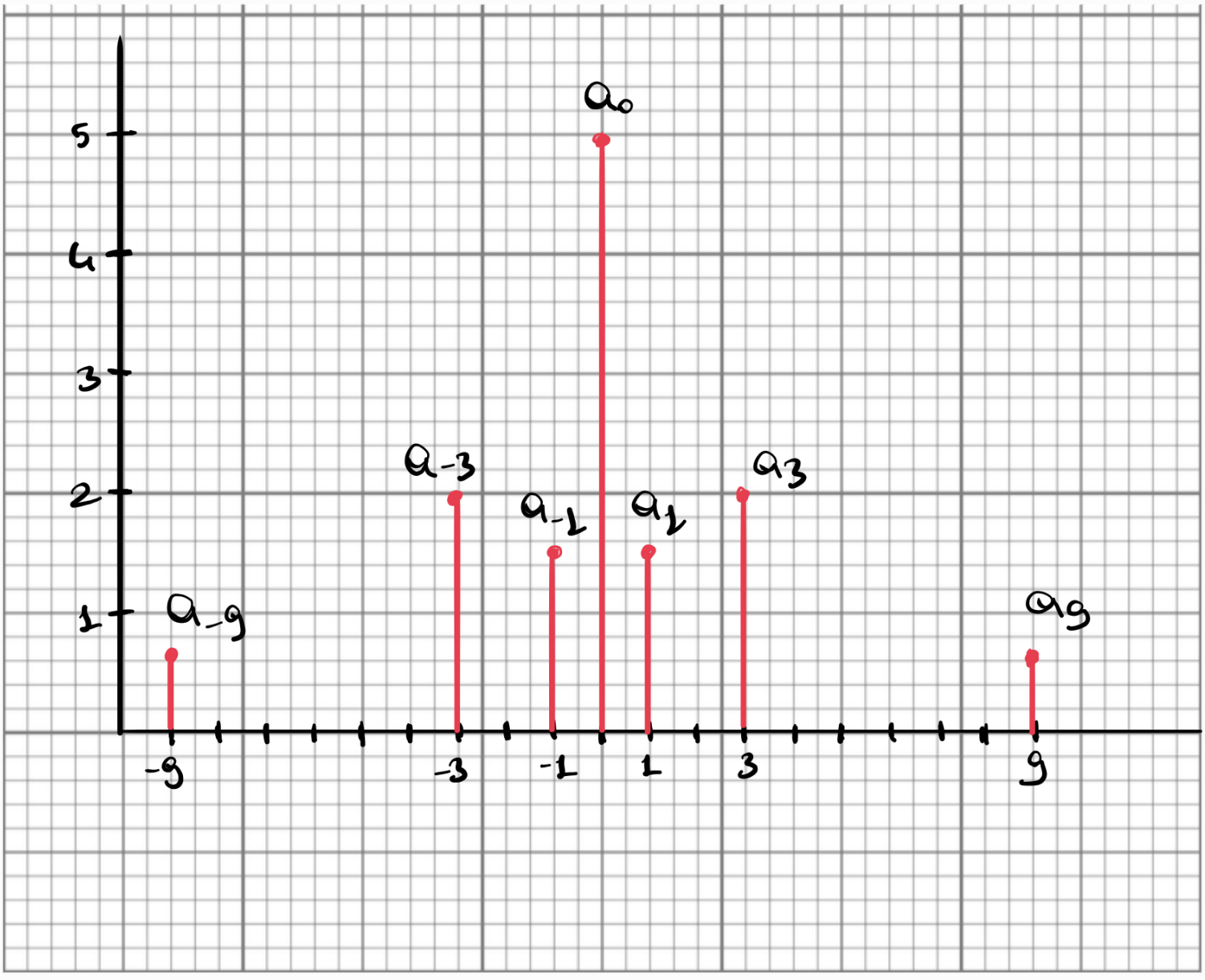
$a_{-1} = \frac{3}{2} e^{j\frac{7\pi}{4}} + \frac{3}{4} e^{-j\frac{\pi}{2}} - 2$

$a_3 = 2e^{j(\pi+5)}$

$a_{-3} = 2e^{j(\pi-5)}$

$a_9 = \frac{3}{4} e^{j\frac{\pi}{2}}$

$a_{-9} = \frac{3}{4} e^{-j\frac{\pi}{2}}$


a_0
5
4
3
a_{-3}
2
a_{-1}
a_1
a_3
1
a_{-9}
a_9
-9
-3
-1
1
3
9

6) we are given $x(t) = \underbrace{\sum_{-\infty}^{\infty} a_k e^{jk\omega_0 t}}_{\text{Fourier Synthesis Formula}}$, $\omega_0 = \frac{2\pi}{T_0}$

a) $y(t) = cx(t - t_0) \Rightarrow y(t) = c\sum_k a_k e^{jk\omega_0(t-t_0)}$

$\Rightarrow y(t) = \sum_k \underbrace{\left(c a_k e^{-jk\omega_0 t_0}\right)}_{y_k} e^{jk\omega_0 t}$

Fourier series coefficients of $y(t)$ are $\boxed{y_k = c a_k e^{-jk\omega_0 t_0}}$

b) $y(t) = \frac{dx(t)}{dt} = \frac{d\left(\sum_k a_k e^{jk\omega_0 t}\right)}{dt} = \sum \underbrace{\left(a_k jk\omega_0\right)}_{y_k} e^{jk\omega_0 t}$

Fourier series coefficients of $y(t)$ are $\boxed{y_k = a_k jk\omega_0}$

7) $x(t) = 5\left|\sin(10\pi t)\right|$, $\phi = 0$ $\Rightarrow T_0 = \frac{2\pi}{\omega_0} = \frac{1}{5}$ sec

instead of using Fourier Analysis formula, we will use one of the Fourier expansions:

$$\left|\sin\theta\right| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{\cos(2n\theta)}{4n^2-1}$$

meaning $x(t) = 5\left|\sin(10\pi t)\right| = \frac{10}{\pi} - \frac{20}{\pi}\sum_{n=1}^{\infty}\frac{\cos(10\pi 2nt)}{4n^2-1}$

in the real Fourier Series Form $x(t) = A_0 + \sum_{k=1}^{\infty} A_k\cos(k\omega_0 t + \phi_k)$

$A_0 = a_0 = \frac{10}{\pi}$.

k is always even since $k = 2n$, meaning for odd coefficients we have $a_{\pm(2n+1)} = 0$ $(n \geq 1)$

and finally $a_{\pm 2n} = \frac{A_{2n}}{2} = \frac{1}{2}\cdot\frac{-20}{\pi(4n^2-1)} = \frac{-10}{\pi(4n^2-1)}$

8) $x(t) = 3\sin\left(120\pi t - \frac{\pi}{3}\right)$

$$x(t) = \frac{3}{2j} e^{-j\frac{\pi}{3}} e^{j120\pi t} - \frac{3}{2j} e^{j\frac{\pi}{3}} e^{-j120\pi t}$$

$$\dots \quad f_{max} = \frac{\omega_{max}}{2\pi} = \frac{120\pi}{2\pi} = 60\,Hz$$

$$f_{Nyquist} = 2 f_{max} = 120\,Hz$$

if $f_s > f_{Nyquist}$, oversampling

if $0 < f_s < f_{max}$ undersampling without folding

if $f_{max} < f_s < 2f_{max}$ // with folding

if $f_s = f_{Nyquist}$ sampling at the Nyquist rate

a) $f_s = 40\,Hz \Rightarrow T_s = \frac{1}{40}\,sec$, $f_s < f_{max}$: undersampling without folding

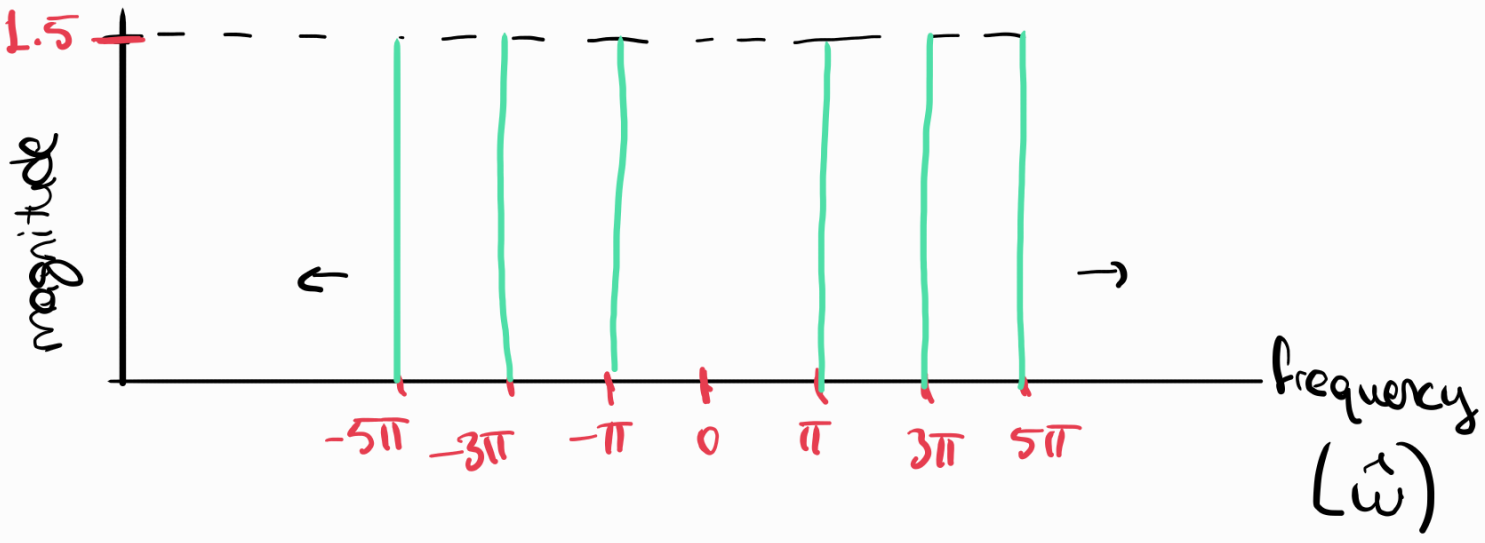
$$x[n] = x(t)\Big|_{t=nT_s} = x(nT_s) = 3\sin\left(3\pi n - \frac{\pi}{3}\right)$$

$$x(nT_s) = \frac{3}{2j} e^{-j\frac{\pi}{3}} e^{j3\pi n} - \frac{3}{2j} e^{j\frac{\pi}{3}} e^{-j3\pi n}$$

$$= \frac{3}{2} e^{-j\frac{5\pi}{6}} e^{j3\pi n} + \frac{3}{2} e^{j\frac{5\pi}{6}} e^{-j3\pi n}$$

$$\left| \frac{3}{2} e^{-j\frac{5\pi}{6}} \right| = 1.5 \quad (\text{for } \hat{\omega} = 3\pi)$$

$$\left| \frac{3}{2} e^{j\frac{5\pi}{6}} \right| = 1.5 \quad (\text{for } \hat{\omega} = -3\pi)$$



$$y(t) = 3\sin\left(40\pi t - \frac{\pi}{3}\right)$$

b) $f_s = 75\ Hz \Rightarrow T_s = \frac{1}{75}\ sec$

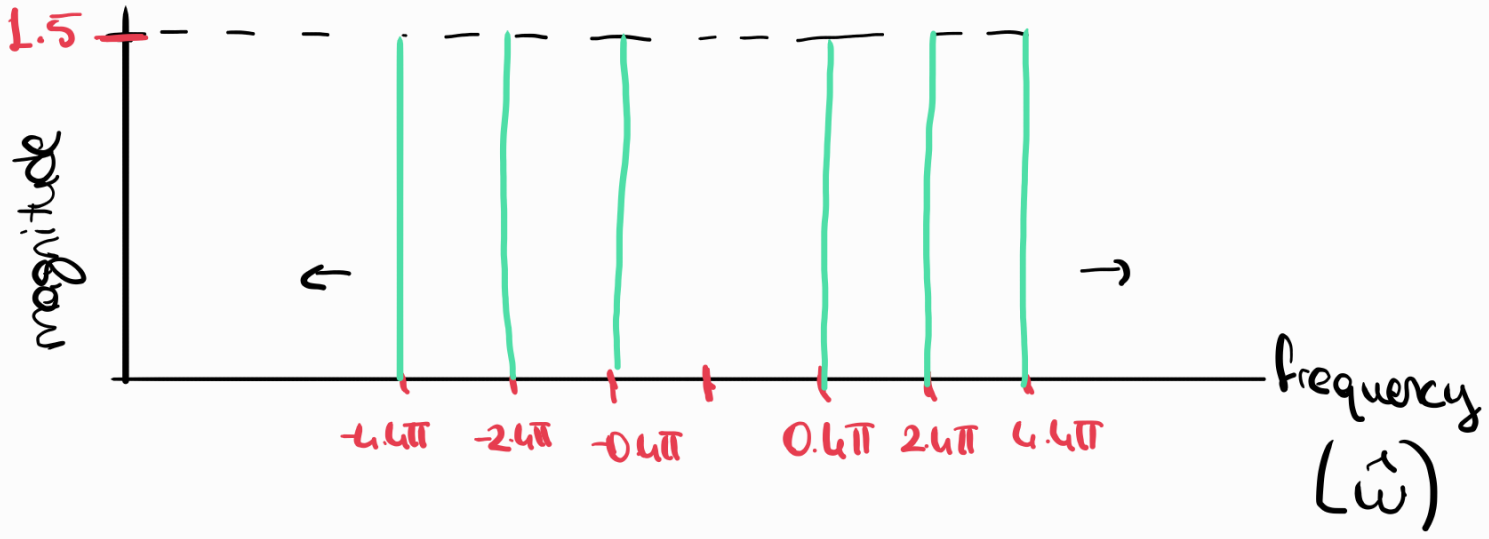
since $f_{max} < f_s < f_{Nyquist}$, undersampling with folding

$$x[n] = x(t)\Big|_{t=nT_s} = x(nT_s) = 3\sin\left(\frac{8\pi}{5}n - \frac{\pi}{3}\right)$$

$$x(nT_s) = \frac{3}{2j}e^{-j\frac{\pi}{3}}e^{j\frac{8\pi}{5}n} - \frac{3}{2j}e^{j\frac{\pi}{3}}e^{-j\frac{8\pi}{5}n}$$

$$= \frac{3}{2}e^{-j\frac{5\pi}{6}}e^{j\frac{8\pi}{5}n} + \frac{3}{2}e^{j\frac{5\pi}{6}}e^{-j\frac{8\pi}{5}n}$$

$$\left|\frac{3}{2}e^{-j\frac{5\pi}{6}}\right| = 1.5 \qquad \left|\frac{3}{2}e^{j\frac{5\pi}{6}}\right| = 1.5$$



$$y(t) = 3\sin(30\pi t - \pi/3)$$

c) $f_s = 180\,Hz \Rightarrow T_s = \frac{1}{180}$ sec

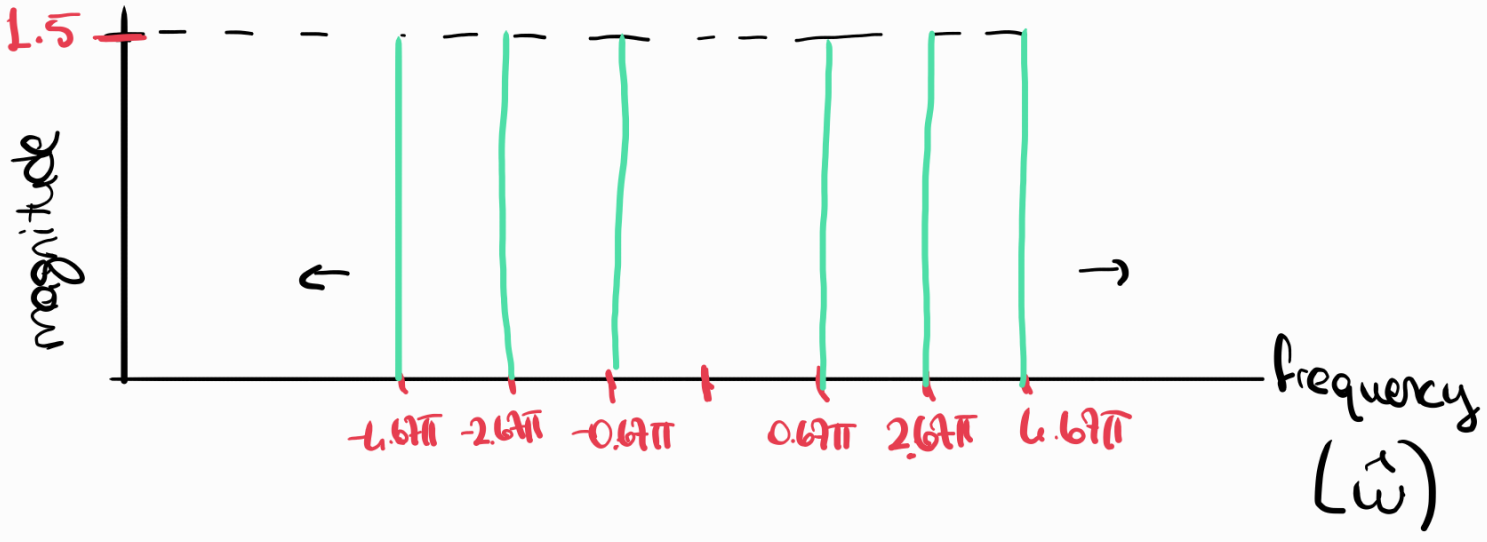
since $f_s > f_{Nyquist}$ oversampled

$$x[n] = x(t)\Big|_{t=nT_s} = x(nT_s) = 3\sin\left(\frac{2\pi}{3}n - \frac{\pi}{3}\right)$$

$$x(nT_s) = \frac{3}{2j}e^{-j\frac{\pi}{3}}e^{j\frac{2\pi}{3}n} - \frac{3}{2j}e^{j\frac{\pi}{3}}e^{-j\frac{2\pi}{3}n}$$

$$= \frac{3}{2}e^{-j\frac{5\pi}{6}}e^{j\frac{2\pi}{3}n} + \frac{3}{2}e^{j\frac{5\pi}{6}}e^{-j\frac{2\pi}{3}n}$$

$$\left|\frac{3}{2}e^{-j\frac{5\pi}{6}}\right| = 1.5 \qquad \left|\frac{3}{2}e^{j\frac{5\pi}{6}}\right| = 1.5$$



$$y(t) = 3\sin\left(120\pi t - \frac{\pi}{3}\right)$$